

ASSIGNMENT -3



K. R. MANGALAM UNIVERSITY

B.Tech CSE (AI & ML)

Section - B

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Course Code: (ENCS352)

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Assignment 3: Simulation and Comparison of RIP and OSPF Routing Protocols

Experiment Objectives

Create a network simulation to demonstrate RIP (Routing Information Protocol) and OSPF (Open Shortest Path First) using Cisco Packet Tracer. Compare the performance, convergence, and efficiency of both routing protocols by simulating data transmission across multiple networks.

Task A: RIP Simulation

Routing

Routing is the process of sending data between networks using routers. Each router uses a **routing table** to decide the path.

RIP (Routing Information Protocol)

- A **distance-vector routing protocol**
- Uses **hop count** to choose the best path
- Max hops = **15** (beyond that unreachable)
- Updates every **30 seconds**

How RIP Works

- Routers share routing information with neighbors
- Each router updates its routing table
- Shortest path (least hops) is selected

Routing Table

Contains:

- Destination network
- Next hop
- Metric (hop count)
- Route type:
 - **C**: directly connected
 - **R**: learned via RIP

Configuration Idea

- Enable RIP on routers
- Add networks to advertise
- Routers automatically exchange routes

Conclusion

RIP is simple and useful for small networks to understand dynamic routing, but not suitable for large-scale networks due to limitations.

Route Table

Router 0

```
Router0
Router>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0
R    20.0.0.0/8 [120/1] via 40.40.40.2, 00:00:28, Serial2/0
R    30.0.0.0/8 [120/2] via 40.40.40.2, 00:00:28, Serial2/0
C    40.0.0.0/8 is directly connected, Serial2/0
R    50.0.0.0/8 [120/1] via 40.40.40.2, 00:00:28, Serial2/0
```

Router 1

```
Router1
Router>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

R    10.0.0.0/8 [120/1] via 40.40.40.1, 00:00:17, Serial2/0
C    20.0.0.0/8 is directly connected, FastEthernet0/0
R    30.0.0.0/8 [120/1] via 50.50.50.2, 00:00:23, Serial3/0
C    40.0.0.0/8 is directly connected, Serial2/0
C    50.0.0.0/8 is directly connected, Serial3/0
```

Router 2

```
Router2
Router>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

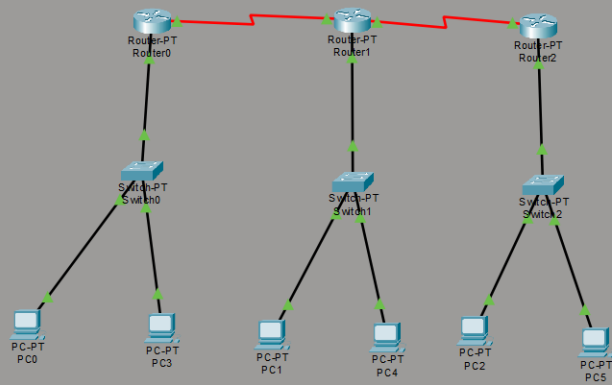
Gateway of last resort is not set

R    10.0.0.0/8 [120/2] via 50.50.50.1, 00:00:20, Serial2/0
R    20.0.0.0/8 [120/1] via 50.50.50.1, 00:00:20, Serial2/0
C    30.0.0.0/8 is directly connected, FastEthernet0/0
R    40.0.0.0/8 [120/1] via 50.50.50.1, 00:00:20, Serial2/0
C    50.0.0.0/8 is directly connected, Serial2/0
```

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 30.30.30.2

Pinging 30.30.30.2 with 32 bytes of data:
Reply from 30.30.30.2: bytes=32 time=20ms TTL=125
Reply from 30.30.30.2: bytes=32 time=48ms TTL=125
Reply from 30.30.30.2: bytes=32 time=86ms TTL=125
Reply from 30.30.30.2: bytes=32 time=52ms TTL=125

Ping statistics for 30.30.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 20ms, Maximum = 86ms, Average = 51ms
```



Task B: OSPF Simulation

Routing

Routing enables communication between different networks. Routers maintain a **routing table** to decide the best path for forwarding data.

OSPF (Open Shortest Path First)

- A **link-state routing protocol**
- Uses **cost (bandwidth-based)** instead of hop count
- Faster and more efficient than RIP
- Works using **areas** (Area 0 = backbone)

How OSPF Works

- Routers discover neighbors using **Hello packets**
- They form adjacency (FULL state)
- Exchange topology information (LSAs)
- Each router calculates shortest path using **Dijkstra's algorithm**

Neighbor Relationship

- Routers must:
 - Be in same area
 - Have matching parameters (hello timers, etc.)
- Successful connection: **State = FULL**

Routing Table

Contains:

- Destination network
- Next hop
- Cost
- Routes learned via OSPF marked as:
 - **O**: OSPF routes
 - **C**: Connected

Shortest Path Selection

- OSPF selects path with **lowest cost**
- Cost depends on **bandwidth**
- If multiple paths exist → best (lowest cost) is chosen

Conclusion

OSPF is a fast and scalable routing protocol that selects the most efficient path using cost, making it suitable for larger and more complex networks compared to RIP.

Check adjacency

Router 0

```
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
50.50.50.1	0	FULL/ -	00:00:32	40.40.40.2	Serial2/0

Router 1

```
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
40.40.40.1	0	FULL/ -	00:00:33	40.40.40.1	Serial2/0
50.50.50.2	0	FULL/ -	00:00:39	50.50.50.2	Serial3/0

Router 2

```
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
50.50.50.1	0	FULL/ -	00:00:38	50.50.50.1	Serial2/0

Check OSPF routes (O) (Routing tables)

Router 0

```
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0
O    20.0.0.0/8 [110/782] via 40.40.40.2, 00:08:33, Serial2/0
O    30.0.0.0/8 [110/846] via 40.40.40.2, 00:07:15, Serial2/0
C    40.0.0.0/8 is directly connected, Serial2/0
O    50.0.0.0/8 [110/845] via 40.40.40.2, 00:08:33, Serial2/0
```

Router 1

```
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

O    10.0.0.0/8 [110/65] via 40.40.40.1, 00:08:59, Serial2/0
C    20.0.0.0/8 is directly connected, FastEthernet0/0
O    30.0.0.0/8 [110/65] via 50.50.50.2, 00:07:51, Serial3/0
C    40.0.0.0/8 is directly connected, Serial2/0
C    50.0.0.0/8 is directly connected, Serial3/0
```

Router 2

```
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

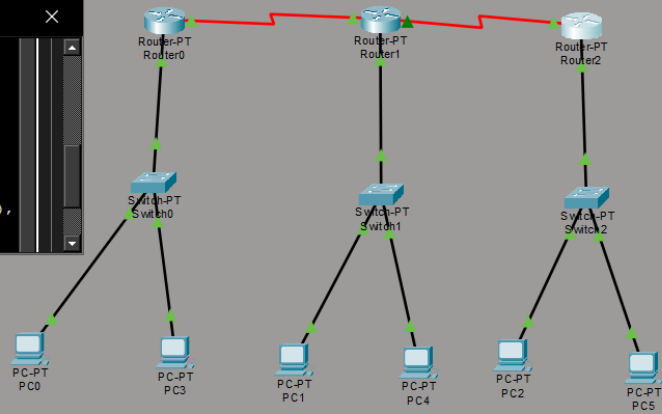
O    10.0.0.0/8 [110/129] via 50.50.50.1, 00:08:15, Serial2/0
O    20.0.0.0/8 [110/65] via 50.50.50.1, 00:08:15, Serial2/0
C    30.0.0.0/8 is directly connected, FastEthernet0/0
O    40.0.0.0/8 [110/128] via 50.50.50.1, 00:08:15, Serial2/0
C    50.0.0.0/8 is directly connected, Serial2/0
```

```
PC0
C:\>ping 30.30.30.2

Pinging 30.30.30.2 with 32 bytes of data:

Reply from 30.30.30.2: bytes=32 time=36ms TTL=125
Reply from 30.30.30.2: bytes=32 time=29ms TTL=125
Reply from 30.30.30.2: bytes=32 time=50ms TTL=125
Reply from 30.30.30.2: bytes=32 time=26ms TTL=125

Ping statistics for 30.30.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 26ms, Maximum = 50ms, Average = 35ms
```



Task C: Comparison of RIP vs OSPF

Feature	RIP	OSPF
Convergence Time	Slow convergence; takes time to update all routers after a change	Fast convergence; quickly updates routing information
Routing Metric	Uses hop count (max 15 hops)	Uses cost based on bandwidth
Efficiency	Sends full routing table every 30 seconds → higher bandwidth usage	Sends updates only when changes occur → more efficient
Scalability	Limited to small networks due to hop limit	Highly scalable; supports large and complex networks using areas

Detailed Explanation

- **Convergence Time:**
RIP updates routes periodically, so changes in the network take time to reflect. OSPF updates immediately when a change occurs, making it much faster and more reliable.
- **Routing Metric:**
RIP simply counts the number of routers (hops), which is not always the best path.
OSPF calculates cost based on bandwidth, so it selects more efficient routes.
- **Efficiency:**
RIP continuously sends complete routing tables, even if no changes occur. OSPF sends only necessary updates, reducing unnecessary traffic.
- **Scalability:**
RIP cannot handle large networks due to its hop limit and slow updates. OSPF is designed for large networks and divides them into areas for better management.

Conclusion

RIP is simple but limited and inefficient. OSPF is faster, more efficient, and suitable for real-world large-scale networks, making it the preferred routing protocol.